Quiz Itself

National University of Computer and Emerging Sciences

Fast School of Computing Spring 2026

CS-1005 Discrete Structures **Quiz # 3 [10 marks]**

Name: _______________ Roll No: _______________ Section: _______________

Time Allocated: 15 Minutes

Question 1: Translation & Negation (5 Marks)

Let S be the set of all students.

Let P be the set of all final projects.

Let C(s) be the predicate "s is a Computer Science major".

Let Submitted(s, p) be the predicate "s has submitted project p".

Consider the following statement:

"For every student s, if s is a Computer Science major, then there will be a project p such that s has submitted p."

a) Translate the statement above into formal symbolic logic using quantifiers ($\forall$, $\exists$) and logical connectives.

b) Formally negate your symbolic statement from part (a). Try to simplify it (only if applicable)

c) Translate your **negated** symbolic statement from part (b) back into a clear, natural English sentence.

Question 2: Truth Values & Counterexamples (5 Marks)

Determine if the following statements are **True** or **False**, and justify your answer

a) Domain: $\mathbb{R}$ (The set of all Real numbers)

Statement: $\forall x \in \mathbb{R}$, if $x^3 > x$, then $x > 0$.

b) Domain: $\mathbb{Z}$ (The set of all Integers)

Statement: $\forall m \in \mathbb{Z}$, $\exists n \in \mathbb{Z}$ such that $2n = m$.

Marking Scheme

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Solution Key: CS-1005 Discrete Structures - Quiz # 3

Question 1: Translation & Negation (5 Marks)

a) Translation to Symbolic Logic $\forall s \in S, (C(s) \rightarrow \exists p \in P, \text{Submitted}(s, p))$

b) Formal Negation and Simplification $\exists s \in S, (C(s) \land \forall p \in P, \neg \text{Submitted}(s, p))$

- **Step-by-Step Derivation:**
  1. Apply negation: $\neg [\forall s \in S, (C(s) \rightarrow \exists p \in P, \text{Submitted}(s, p))]$
  2. Negate universal quantifier: $\exists s \in S, \neg (C(s) \rightarrow \exists p \in P, \text{Submitted}(s, p))$
  3. Negate implication ($P \rightarrow Q \equiv P \land \neg Q$): $\exists s \in S, (C(s) \land \neg (\exists p \in P, \text{Submitted}(s, p)))$
  4. Negate existential quantifier: $\exists s \in S, (C(s) \land \forall p \in P, \neg \text{Submitted}(s, p))$

c) English Translation of the Negated Statement (1.5 Marks) "There exists a student who is a Computer Science major, but has not submitted any final project." *OR SIMILAR*

Question 2: Truth Values & Counterexamples (5 Marks)

a) Domain: $\mathbb{R}$ | Statement: $\forall x \in \mathbb{R}$, if $x^3 > x$, then $x > 0$. (2.5 Marks)

- **Answer:** False
- **Justification:** To disprove the implication, we need a counterexample where the premise is True, but the conclusion is False. Let $x = -0.5$. The premise $x^3 > x$ evaluates to $(-0.5)^3 > -0.5$, which is $-0.125 > -0.5$. This is **True**. However, the conclusion $x > 0$ evaluates to $-0.5 > 0$, which is **False**. Since True $\rightarrow$ False is logically False, the statement does not hold for all real numbers.

b) Domain: $\mathbb{Z}$ | Statement: $\forall m \in \mathbb{Z}, \exists n \in \mathbb{Z}$ such that $2n = m$. (2.5 Marks)

- **Answer:** False
- **Justification:** The statement claims that every integer m is even (can be written as 2n where n is an integer). We can disprove this using any odd integer. Let $m = 3$. The statement requires an integer n such that $2n = 3$. Solving for n gives $n = 1.5$. Since 1.5 is not an integer ($1.5 \notin \mathbb{Z}$), no such integer n exists for $m = 3$.